

NORTH ADAMS, MA, USA

WMO: 725075

Lat: 42.697N

Lon: 73.170W

Elev: 640

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54768

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.1	(c) 3.0	(d) -12.6	(e) 2.8	(f) 0.4	(g) -8.3	(h) 3.6	(i) 5.4	(j) 26.2	(k) 24.3	(l) 24.3	(m) 24.7	(n) 6.1	(o) 270	(p) 0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	86.8	71.6	84.1	69.9	81.8	68.9	74.2	82.8	72.6	80.7	71.2	78.7	6.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	120.4	78.7	70.1	113.5	76.9	68.5	107.1	75.6	38.0	82.6	36.6	80.3	35.3	78.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.8	(b) 18.9	(c) 16.7	(e) -9.8	(f) 91.1	(g) 5.7	(h) 2.0	(i) -13.9	(j) 92.6	(k) -17.2	(l) 93.7	(m) -20.4	(n) 94.8	(o) -24.5	(p) 96.3		
(4) 21.8	(5) 18.9	(6) 16.7	(7) -9.8	(8) 91.1	(9) 5.7	(10) 2.0	(11) -13.9	(12) 92.6	(13) -17.2	(14) 93.7	(15) -20.4	(16) 94.8	(17) -24.5	(18) 96.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	23.2	26.3	33.7	45.9	56.9	65.0	69.3	67.8	61.2	49.9	39.4	30.0
	DBStd	18.09	11.70	10.01	9.91	8.68	7.64	6.25	5.03	5.18	6.90	8.19	8.92	9.69
	HDD50	3277	831	664	514	180	20	0	0	0	3	107	336	622
	HDD65	6798	1295	1083	971	577	270	76	16	31	154	472	768	1085
	CDD50	2367	1	1	8	56	235	451	600	553	339	103	18	2
	CDD65	413	0	0	0	3	20	77	151	119	40	3	0	0
	CDH74	3791	0	0	11	78	326	764	1358	911	320	23	0	0
	CDH80	926	0	0	2	20	77	206	365	187	68	1	0	0
	WSAvg	6.0	8.0	8.2	7.9	7.3	5.6	4.2	3.3	3.2	3.8	5.7	6.9	8.0
	PrecAvg	48.7	3.4	2.8	4.1	3.7	4.3	4.3	4.6	4.4	4.6	4.4	4.1	3.9
(15) Precipitation	PrecMax	61.2	5.7	7.3	6.5	6.3	8.0	7.6	9.2	9.5	7.5	12.2	6.1	6.7
	PrecMin	36.0	1.8	1.6	1.4	1.2	1.1	0.6	1.9	1.8	1.9	1.2	2.5	1.4
	PrecStd	6.7	1.1	1.2	1.2	1.3	1.7	1.8	1.9	2.2	1.8	2.5	1.1	1.4
	PrecStd	6.7	1.1	1.2	1.2	1.3	1.7	1.8	1.9	2.2	1.8	2.5	1.1	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	58.1	59.7	73.7	82.7	86.2	89.8	90.3	88.4	86.4	77.1	69.5	60.9	
	0.4% MCWB	53.6	51.6	57.0	62.2	67.5	72.9	73.9	74.0	72.3	64.5	61.0	54.7	
	2% DB	48.5	51.2	61.3	74.2	82.0	85.3	86.6	84.3	81.3	72.3	63.7	54.0	
	2% MCWB	43.7	44.1	50.6	56.8	65.3	69.8	72.1	71.2	69.2	62.7	56.0	49.9	
	5% DB	43.1	44.6	54.3	68.2	77.5	81.8	84.0	81.8	77.7	68.2	58.9	48.0	
	5% MCWB	39.4	38.5	45.7	54.3	62.5	68.4	70.4	70.1	67.2	60.5	52.3	44.5	
	10% DB	38.6	39.4	48.4	62.8	72.8	78.7	81.5	79.5	73.5	63.7	54.2	43.3	
	10% MCWB	35.2	34.9	41.6	50.8	60.3	66.2	69.2	68.4	64.9	56.9	48.8	39.1	
	0.4% WB	54.5	53.9	58.0	65.3	70.4	75.2	76.5	76.0	74.0	68.5	62.4	56.4	
	0.4% MCDB	57.2	59.7	70.4	75.5	82.1	85.4	86.3	84.3	82.7	73.8	67.6	60.4	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	45.5	44.2	52.1	60.2	67.7	72.7	74.2	73.8	71.5	65.1	57.3	50.0	
	2% MCDB	47.7	49.3	59.3	71.3	78.0	81.9	82.9	81.3	79.1	70.3	62.3	53.5	
	5% WB	39.6	39.0	46.4	56.5	65.3	70.8	72.6	72.1	69.1	61.4	53.5	45.0	
	5% MCDB	43.1	43.5	52.6	65.8	74.5	78.9	80.8	79.0	75.2	66.8	57.8	48.2	
	10% WB	35.6	35.7	42.2	51.9	62.6	68.4	71.0	70.4	66.8	57.8	49.1	39.9	
	10% MCDB	38.4	39.6	48.1	60.5	70.6	75.5	78.6	76.5	72.2	62.8	53.9	42.9	
(35) Mean Daily Temperature Range	MDBR	16.3	17.9	18.4	21.7	22.4	21.2	21.3	20.9	21.4	19.4	17.0	14.3	
	5% DB	19.2	23.6	27.7	32.4	30.1	26.5	24.2	22.8	25.1	26.0	24.2	19.8	
	5% WB	17.0	17.8	17.6	17.7	15.4	13.4	11.6	11.8	14.0	16.3	17.6	17.5	
	MCDBR	18.7	22.2	24.8	27.2	26.2	22.9	21.0	19.4	21.2	22.6	20.2	19.2	
	MCWBR	17.2	17.7	17.5	16.9	14.5	12.5	11.0	10.9	12.5	15.8	16.7	17.6	
	MCWBR	17.2	17.7	17.5	16.9	14.5	12.5	11.0	10.9	12.5	15.8	16.7	17.6	
(40) Clear-Sky Solar Irradiance	taub	0.294	0.311	0.330	0.354	0.388	0.434	0.453	0.426	0.369	0.345	0.310	0.289	
	taud	2.380	2.326	2.383	2.402	2.342	2.236	2.194	2.253	2.396	2.474	2.522	2.480	
	Ebn at Noon	268	281	290	290	281	267	260	264	273	267	263	262	
	Edh at Noon	25	31	34	36	39	43	45	41	33	27	22	21	
	RadAvg	514	793	1131	1437	1626	1760	1834	1638	1333	837	572	410	
(45) All-Sky Solar Radiation	RadStd	42	75	89	158	176	159	124	86	108	89	58	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+79

Nomenclature: See separate page